

DrGPharma One-Pot System — Developer Implementation Spec

Purpose

This document explains how to implement the DrGPharma **One-Pot System** in the software. The system uses a single ring-fenced financial pot for Dr G. Patient payments, stock purchases, pharmacy reimbursements, VAT positions, refunds, and accumulated commission are tracked inside one controlled ledger.

The software must show, in real time, how much cash is available, how much is tied to stock, what VAT is reclaimable or owed, what profit has been earned, and whether any money is temporarily restricted for HMRC or pending stock purchase.

The goal is simple: **Dr G's money, stock, margin, VAT benefit, and reimbursements must be visible and auditable in one place.**

Core Concept

The system should maintain one financial pot per partner/brand, for example:

Partner Pot: DrGPharma

This pot is not just a bank balance. It is a ledger-backed balance containing several calculated sub-balances:

- **Cash balance** — actual available cash in the pot.
- **Stock value** — value of stock purchased for Dr G and still unsold.
- **VAT position** — running VAT reclaim or VAT payable position.
- **Pending VAT refund** — VAT refund expected from HMRC but not yet received.
- **Restricted VAT payable** — VAT owed to HMRC and therefore not spendable.
- **Earned commission / profit** — cumulative Dr G profit generated by orders.
- **Working float** — optional small buffer used in pre-order mode to cover timing shortfalls.

All movement must be recorded as ledger entries. Do not calculate pot value only from order totals. The ledger must be the source of truth.

Recommended Data Model

1. Partner Pot

Create a `partner_pots` table or equivalent model.

Suggested fields:

```
id
partner_id
name
currency
cash_balance
stock_value
vat_position
pending_vat_refund
restricted_vat_payable
earned_profit
working_float
available_to_spend
status
created_at
updated_at
```

Important calculated fields:

```
available_to_spend = cash_balance - restricted_vat_payable - pending_purchase_commitments
true_pot_value = cash_balance + stock_value + pending_vat_refund - restricted_vat_payable
```

Do not allow the system to spend restricted VAT payable.

2. Ledger Entries

Create a `pot_ledger_entries` table.

Suggested fields:

```
id
partner_pot_id
order_id nullable
stock_purchase_id nullable
vat_return_id nullable
type
amount
vat_amount
cash_delta
stock_delta
vat_position_delta
profit_delta
restricted_delta
description
metadata_json
created_by
created_at
```

Suggested entry types:

```
INITIAL_FLOAT
PATIENT_PAYMENT_RECEIVED
CARD_FEE_DEDUCTED
STOCK_PURCHASE
STOCK_ALLOCATED_TO_ORDER
PACKAGING_REIMBURSEMENT
DELIVERY_REIMBURSEMENT
VAT_INPUT_RECORDED
VAT_OUTPUT_RECORDED
VAT_REFUND_EXPECTED
VAT_REFUND_RECEIVED
VAT_PAYABLE_RESTRICTED
VAT_PAYABLE_RELEASED
COMMISSION_EARNED
MANUAL_ADJUSTMENT
REFUND_TO_PATIENT
ORDER_CANCELLED
STOCK_WRITE_OFF
```

Every financial event must produce one or more ledger entries.

3. Orders

Each order should track whether it is fulfilled from shelf stock or pre-order stock.

Suggested fields:

```
id
partner_id
partner_pot_id
customer_id
order_number
fulfilment_mode ENUM('SHELF_STOCK', 'PRE_ORDER', 'HYBRID')
status
subtotal_inc_vat
shipping_inc_vat
card_fee
net_settlement
stock_cost_inc_vat
stock_cost_ex_vat
input_vat
output_vat
packaging_cost
delivery_cost
cash_impact
vat_position_impact
profit
created_at
paid_at
stock_ordered_at
stock_received_at
dispatched_at
cancelled_at
```

4. Order Items

Each order item should know whether it is POM or non-POM.

Suggested fields:

```
id
order_id
product_id
product_name
quantity
sale_price_inc_vat
sale_price_ex_vat
sale_vat_rate
sale_vat_amount
cost_price_inc_vat
cost_price_ex_vat
input_vat_amount
is_pom
is_zero_rated_on_prescription
fulfilment_source ENUM('SHELF', 'WHOLESALE_PRE_ORDER')
```

Product VAT Rules

The software must support product-specific VAT behaviour.

Azzalure Example

Azzalure is treated as a POM and is zero-rated when dispensed against a valid prescription.

Example:

```
Sale price to patient: £145.00
Output VAT charged to patient: £0.00
Purchase cost inc-VAT: £152.39
Input VAT paid on purchase: £25.40
VAT position impact: +£25.40 reclaimable from HMRC
```

This means Azzalure may look cash-negative at order level, but profitable after VAT recovery.

Restylane Refyne Example

Restylane Refyne is treated as standard-rated.

Example:

```
Sale price inc-VAT: £92.38
Sale price ex-VAT: £76.98
Output VAT: £15.40
Purchase cost inc-VAT: £83.00
Purchase cost ex-VAT: £69.17
Input VAT: £13.83
VAT position impact: -£1.57 payable to HMRC
Product margin ex-VAT: £7.81
Net contribution after VAT payable: £6.24
```

One-Pot Financial Formula

For every order, calculate two separate values:

1. Immediate Cash Impact

This is the cash movement before HMRC VAT refunds are received.

```
cash impact = patient payment_received
              - card_fee
              - packaging cost
              - delivery cost
              - stock_cost_inc_vat_if_purchased_for_this_order
```

In shelf-stock mode, stock was already purchased, so the order may show a large positive cash inflow, while stock value decreases.

In pre-order mode, stock is purchased after payment, so the stock cost must be deducted from the patient payment.

2. True Profit / Commission

This is the real commercial result after VAT position is included.

```
true profit = patient payment_received
              - card_fee
              - packaging cost
              - delivery cost
              - stock cost inc vat
              + vat_position_impact
```

Where:

```
vat_position_impact = input_vat_reclaimable - output_vat_payable
```

For POM-heavy orders, this is usually positive because input VAT is reclaimed and output VAT is zero.

Example: Mixed Order

Order contains:

```
Azzalure sale: £145.00
Restylane Refyne sale: £92.38
P&P charged: £12.14
Total patient pays: £249.52
```

Costs:

```
Card fee: £2.90
Packaging + delivery: £12.14
Azzalure cost inc-VAT: £152.39
Refyne cost inc-VAT: £83.00
Total stock cost: £235.39
```

VAT:

```
Azzalure VAT reclaim: +£25.40
Refyne VAT payable: -£1.57
Net VAT position: +£23.83
```

Pre-order cash impact:

```
£249.52 - £2.90 - £12.14 - £235.39 = -£0.91
```

True profit:

```
-£0.91 + £23.83 = £22.92
```

The system must show both figures separately:

```
Immediate cash impact: -£0.91
VAT position impact: +£23.83
True profit: £22.92
```

Fulfilment Modes

Mode 1: Shelf Stock

Flow:

1. Pot buys stock in advance.
2. Stock value increases, cash decreases.
3. Patient places order and pays.
4. Order is fulfilled from existing stock.
5. Stock value decreases.
6. Patient payment enters pot after card fee.
7. Packaging/delivery reimbursement is paid out.
8. VAT position is updated.
9. Profit is calculated.

Ledger entries:

```
STOCK PURCHASE: cash down, stock value up, input VAT recorded
PATIENT_PAYMENT_RECEIVED: cash up
CARD FEE DEDUCTED: cash down
STOCK ALLOCATED TO ORDER: stock value down
PACKAGING REIMBURSEMENT: cash down
DELIVERY REIMBURSEMENT: cash down
VAT_OUTPUT_RECORDED / VAT_INPUT_RECORDED
COMMISSION_EARNED
```

Mode 2: Pre-Order

Flow:

1. Patient places order and pays upfront.
2. Net payment settles into pot after card fee.
3. System creates wholesaler purchase order.
4. Pot pays wholesaler for exact stock required.
5. Stock arrives.
6. Stock is immediately allocated to the paid order.
7. Order is dispatched.
8. Packaging/delivery reimbursement is paid out.
9. VAT position is updated.
10. Profit is calculated.

Ledger entries:

```
PATIENT PAYMENT RECEIVED: cash up
CARD FEE DEDUCTED: cash down
STOCK PURCHASE: cash down, temporary stock/pending stock up
STOCK ALLOCATED TO ORDER: stock/pending stock down
PACKAGING REIMBURSEMENT: cash down
DELIVERY REIMBURSEMENT: cash down
VAT INPUT RECORDED
VAT OUTPUT RECORDED
COMMISSION_EARNED
```

Important: in pre-order mode, the pot may show a small temporary cash shortfall for Azzalure-heavy orders until VAT refunds arrive. The software should support a configurable working float.

Mode 3: Hybrid

Flow:

1. Fast-moving products can be held as shelf stock.
2. Slow-moving or expensive products are pre-ordered after patient payment.
3. A single order may contain both shelf-stock and pre-order items.
4. The pot must track stock movement and pre-order purchase movement separately.

Example:

```
Azzalure fulfilled from shelf.
Refyne ordered from wholesaler after payment.
```

The order should not dispatch until all items are ready.

Order State Machine

Recommended statuses:

```
DRAFT
AWAITING_PAYMENT
PAID
PAYMENT SETTLED
STOCK CHECK
WHOLESALE_ORDER_REQUIRED
WHOLESALE_ORDER_PLACED
AWAITING_STOCK
STOCK RECEIVED
READY TO DISPATCH
DISPATCHED
COMPLETED
CANCELLED
REFUNDED
FAILED
```

For shelf stock:

```
PAID → PAYMENT_SETTLED → STOCK_CHECK → READY_TO_DISPATCH → DISPATCHED → COMPLETED
```

For pre-order:

```
PAID → PAYMENT_SETTLED → WHOLESALE_ORDER_REQUIRED → WHOLESALE_ORDER_PLACED → AWAITING_STOCK → STOCK_RECEIVED → READY_TO_DISPATCH → DISPATCHED → COMPLETED
```

VAT Return Handling

The system should maintain a rolling VAT ledger.

Each order updates the VAT position immediately, but the cash effect occurs only when HMRC refund/payment is processed.

Suggested VAT fields per order item:

```
input_vat_paid
output_vat_charged
vat_position_delta = input_vat_paid - output_vat_charged
```

If `vat_position_delta` is positive, HMRC owes the pot.

If `vat_position_delta` is negative, the pot owes HMRC.

VAT Refund Cycle

Recommended implementation:

1. Accumulate VAT position by VAT period.
2. At period close, freeze VAT period.
3. Mark reclaim amount as pending VAT refund.
4. When HMRC pays refund, create `VAT_REFUND_RECEIVED` ledger entry.
5. Increase cash balance and reduce pending VAT refund.

For VAT owed:

1. Ring-fence payable amount immediately.
2. Increase restricted vat payable.
3. Reduce available to spend.
4. When VAT is paid to HMRC, reduce cash and `restricted_vat_payable`.

Ring-Fencing Rules

The software must prevent accidental spending of money that belongs to HMRC or has already been committed.

Restrict VAT Payable

If output VAT exceeds input VAT for a period or order:

```
restricted_vat_payable += amount_owed
available_to_spend -= amount_owed
```

Restrict Pending Purchase Commitments

If a wholesaler order has been placed but not yet settled:

```
pending_purchase_commitments += expected_purchase_cost
available_to_spend -= expected_purchase_cost
```

Prevent Restock If Insufficient Available Cash

Before any stock purchase:

```
if available_to_spend < purchase_total:
    block_purchase_or_request_top-up / working float approval
```

Dashboard Requirements

The admin dashboard should show the one-pot system clearly.

Main Pot Summary

Display:

Cash balance
Available to spend
Stock value
Pending VAT refund
VAT payable restricted
Earned profit
Working float used
True pot value

Order-Level Breakdown

For every order, show:

Patient paid
Card fee
Packaging/delivery cost
Stock cost inc-VAT
Input VAT
Output VAT
VAT position impact
Immediate cash impact
True profit
Fulfilment mode

Stock View

Show:

Stock held for Dr G
Stock value inc-VAT
Units available
Units allocated
Units pending from wholesaler
Average cost
Expiry date if applicable

VAT View

Show:

Current VAT period
Input VAT reclaimable
Output VAT payable
Net VAT position
Expected refund/payment date
Pending refunds
Refunds received
VAT payments made

Alerts

Trigger alerts for:

Available cash below threshold
VAT payable restricted
Pre-order purchase awaiting stock
Wholesaler order delayed
Pot temporarily negative
Working float used
Stock below reorder threshold
Stock expiry approaching
Refund/cancellation requiring reversal

Handling Refunds and Cancellations

Cancel Before Wholesaler Order

If patient cancels after payment but before stock is ordered:

Refund patient.
Reverse patient payment.
Reverse card fee if applicable or record non-refundable fee.
No stock or VAT input impact.

Cancel After Wholesaler Order But Before Dispatch

If stock has been ordered but not dispatched:

Refund patient.
Stock remains in Dr G pot inventory unless returned to wholesaler.
If returned, reverse stock purchase and VAT input.
If retained, cash remains reduced but stock value remains.

Refund After Dispatch

If patient refund occurs after dispatch:

Reverse revenue.
Adjust output VAT.
Stock is not automatically restored unless product is returnable and resaleable.
Profit must be recalculated.
Manual review required for POM items.

Wholesaler Integration Requirements

For pre-order mode, the software should support wholesaler purchase orders.

Suggested fields:

```
purchase_order_id
partner_pot_id
order_id
supplier_id
status
items
expected_total_inc_vat
expected_input_vat
placed_at
received_at
cancelled_at
invoice_reference
```

Statuses:

```
DRAFT
PLACED
CONFIRMED
PARTIALLY_RECEIVED
RECEIVED
CANCELLED
FAILED
```

When a patient order requires pre-order items, the system should automatically create a purchase order or queue it for admin approval.

Example Ledger Flow: Mixed Pre-Order

Patient pays £249.52 for Azzalure + Refyne + P&P.

1. PATIENT PAYMENT RECEIVED
cash_delta: +249.52
2. CARD FEE DEDUCTED
cash_delta: -2.90
3. STOCK PURCHASE
cash_delta: -235.39
stock_delta: +235.39
vat_position_delta: +39.23 input VAT recorded before output split if tracked gross
4. STOCK ALLOCATED TO ORDER
stock_delta: -235.39
5. PACKAGING REIMBURSEMENT / DELIVERY_REIMBURSEMENT
cash_delta: -12.14
6. VAT OUTPUT RECORDED
vat_position_delta: -15.40 for Refyne output VAT
7. VAT INPUT RECORDED
vat position delta: +25.40 Azzalure input VAT reclaim
vat position delta: +13.83 Refyne input VAT reclaim
net VAT position impact: +23.83
8. COMMISSION EARNED
profit_delta: +22.92

Depending on implementation, VAT input/output can be recorded as separate entries or as one net VAT entry. Separate entries are preferred for auditability.

Permissions and Controls

Only authorised admins should be able to:

- Manually adjust pot balances
- Override VAT classification
- Approve negative pot/working float usage
- Mark VAT refund as received
- Mark VAT payable as paid
- Write off stock
- Change cost prices after order completion

All overrides must be logged with:

- user_id
- reason
- before_value
- after_value
- timestamp

Developer Acceptance Criteria

The implementation is complete when:

1. Every order creates auditable ledger entries.
2. The pot shows cash, stock, VAT, restricted, and profit balances separately.
3. Shelf-stock, pre-order, and hybrid fulfilment modes are supported.
4. POM zero-rated and standard-rated product VAT logic works correctly.
5. VAT refunds increase cash only when marked as received.
6. VAT payable is ring-fenced and cannot be spent.
7. The dashboard shows immediate cash impact and true profit separately.
8. Refunds and cancellations reverse the correct ledger entries.
9. The system can explain why the pot is positive, negative, restricted, or awaiting VAT refund.
10. All manual adjustments are audited.

Key Business Rules To Remember

The software must not treat patient payment as pure profit.

Patient payment first funds:

1. Card fee
 2. Stock purchase
 3. Packaging/delivery reimbursement
 4. VAT payable if applicable

Only then does profit emerge.

For Azzalure, profit depends on VAT recovery.

For Refyne, profit exists as product margin, but some VAT is owed to HMRC.

The one-pot system works because all of these are tracked together in one ledger-backed pot.